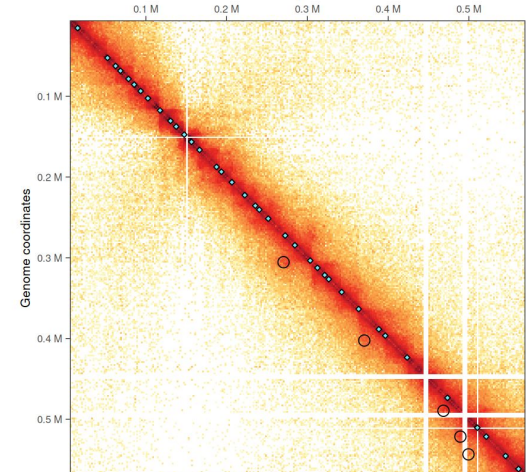
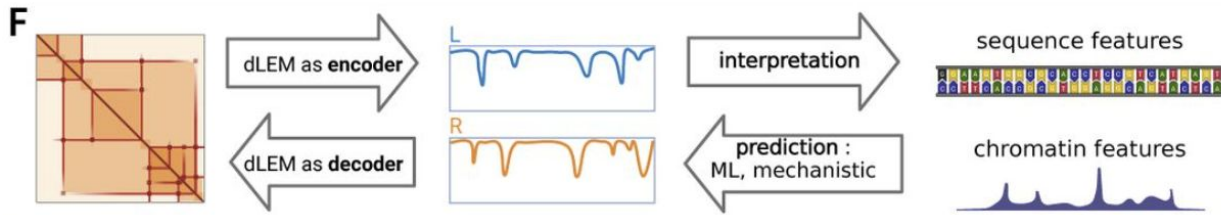


Research Update

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dLEM

- A model which can help us relate 3D structure of the genome to its sequence.
 - Describes a Hi-C contact frequency matrix in two sets of parameters which represent the action of cohesin protein at different genomic loci.
 - The set of parameters represent how quickly each arm of cohesin travels along a strand of DNA.
 - Positions where cohesin slows down is how we identify loops.



My Question

- What is the relationship between fit parameters and input Hi-C data?
 - Hi-C data can be noisy, what is the relationship between noise and model performance?
 - How much noise vs signal is encoded in the parameters?
 - Are parameter sets unique to Hi-C matrices?

Current Goals

- Create/find a dataset that dLEM performs very well on (low mse between input matrix and predicted matrix).
 - To isolate the effect of adding noise, need noise “free” starting place.
- Add noise to input matrix and measure performance changes.

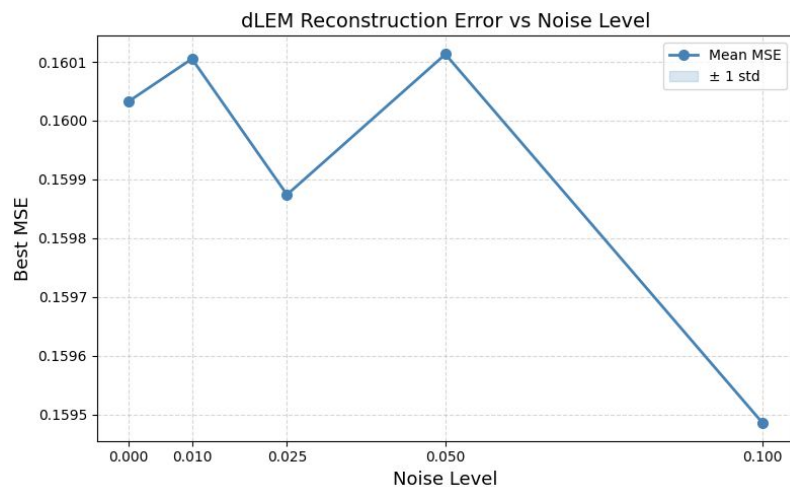
Drosophila data

- Train dLEM on chr3R (.mcool), 20050 bins, 1600 bp/bin (encode)
 - `step=0080 loss=32.176052 corr=0.5884 mse=0.197711`
- p_left and p_right → a synthetic matrix (dlem decode) → .cool
- Train dLEM on synthetic.cool, compare mse between synthetic matrix and output

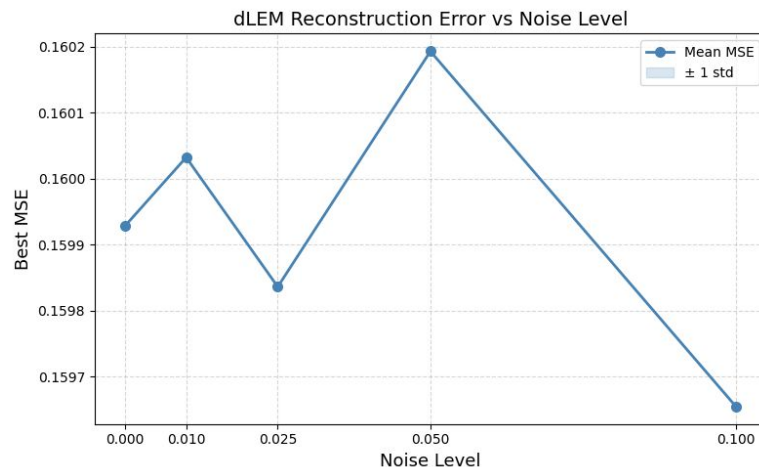
Problem?

Synthetic.cool pixel values from generate_dlem_prediction() (dlem as decoder) are on a different scale than normal .cool file

Maybe rescaling would give better mse.



Lr: 0.001



Lr: 0.0001

step=0300 loss=15.046777 corr=0.3058 mse=0.160033

Fetch_band() on synthetic vs .mcool

cur_pred shape: (170, 20050)

cur_pred mean: 1.0164

cur_pred max: 6.9939

real_band shape: (170, 20050)

real_band mean: 0.0019

real_band max: 0.6408

Research Update

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Where I am at

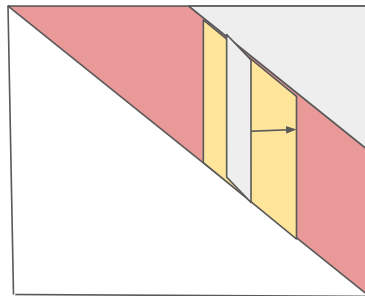
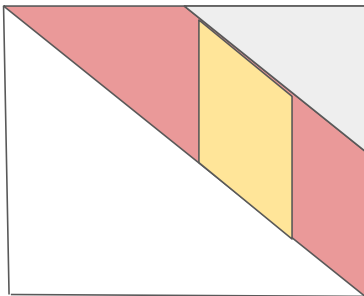
- I want low mse between input matrix and generated matrix for my baseline
- Found the pixels values were on a different scale in my synthetic and real .cool inputs
- Trace through dlem run to find out why

Dlem trace

Input: .mcool file, pairwise bins and their count (contact frequency) →

`fetch_band()` selects a band of n diagonals in a specific region of the chromatin, replaces values below a cut off with a local average, replaces nans with row mean →

`fit_band_row_sliding()` slides a fixed window size over band, calculates row mean (change in contact frequency as distance between bins increase) and fits decay model to row means at each window, returns median param vals and fits across all windows. The natural log is taken of row means (target) and model output.



Dlem trace

`train_dlem()` selects target band, initializes left and right params as 1s →

`forward_fn()` and `loss_fn()` creates predicted band from left and right params, fits left and right params with target and predicted band →

`Jax_forward generate()` builds band with fit left and right params →

`normalize_expected_observed()` returns band with $\text{pixel} = \log((\text{pixel} + 1e-6) / (\text{row_mean} + 1e-6))$

normalized generated band is compared to normalized target band at the end of a training step to collect mse and other metrics

- Final prediction is not normalized, only the predicted band for fitting left and right params
- Final mse is not calculated from the final prediction, it is calculated from the normalized predicted bands and normalized target
- Going to review/compare the shape and values of the final prediction on synthetic and real .mcool files, as well as .cool file
- Maybe use corr as target metric, measure structure

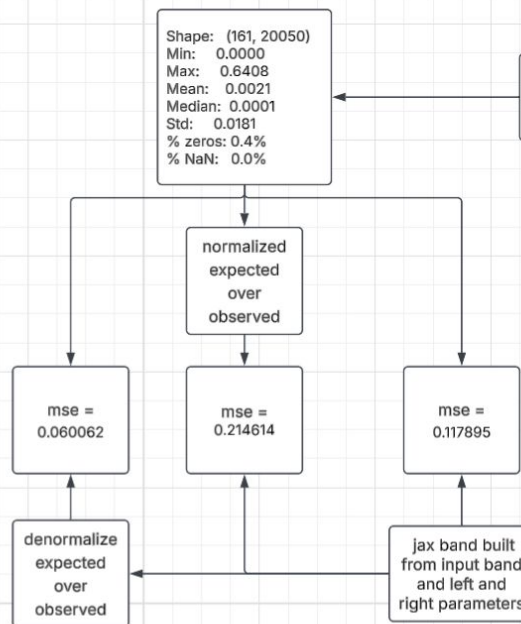
Research Update

7/7

Where I am at

- I need an input matrix and generated matrix for my baseline
- Wrote a function to reverse normalize_expected_observed
- Collected mse between input and dlem encode training band and dlem decode predicted band

Encode



.matrix.fetch_band(region)

Glamda.mcool and

.matrix.fetch_band(region)

call dlem cli, (chr3R)

training metrics, (dlem_results.pkl)
best_mse = 0.19229
(pred generated with foward_fn)

dlem_pred.cool file
(pred matrix)

.matrix.fetch_band(region)

Shape: (170, 20050)
Min: 0.0000
Max: 0.6408
Mean: 0.0019
Median: 0.0001
Std: 0.0176
% zeros: 0.4%
% NaN: 0.0%

mse
=0.09928098

Shape: (170, 20050)
Min: 0.0012
Max: 4.5330
Mean: 0.1887
Median: 0.0639
Std: 0.2593
% zeros: 0.0%
% NaN: 0.0%

Decode

- Input and dlem decode prediction (170,20050)
 - MSE = .09928098
- Input and dlem encode prediction (161, 20050)
 - MSE = 0.117895
- Input and denormalized dlem encode (161, 20050)
 - 0.060062
- Normalized input and dlem encode prediction (161, 20050)
 - 0.214614

- Check if correlation values follow the same pattern as mse
- Run decode prediction back through to reduce mse, and check if structure is preserved
- Use the dlems visualization functions for each of the band